Rules of the Road on the Moon and Mars: Traffic Rules for Surface Robots of Different Owners, Checked and Priced

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Roatify Research · Preprint, October 2026 · Code, data and pre-registration: github.com/azizjonrahimov/Lunar-Haul · Live site: azizjonrahimov.github.io/Lunar-Haul

Abstract. Rovers and robots of several companies will soon share the roads and landing zones of a lunar base, and later a Mars base, with no traffic rules written for them; policy documents ask for coordination but give none. We propose rules for vehicles of different owners: separation from braking physics, exclusive one-lane reservations with an exit bay, timed landing closures, crew keep-clear zones and a lost-contact rule, enforced by a safety program on each vehicle. We checked them three ways on one simulated base: an exhaustive search of a discrete model, 274,580 encounters of twelve types, and 650 runs of a logistics model that prices them in cargo delay at one to eight times today's fleet, then ran the encounters and the two road modes again at Mars gravity with a Mars Earth link. The safety program allowed no violation in eleven of twelve encounter types; the exception, a vehicle stopping dead in a one-lane segment, defeats a gap rule derived for a braking leader; the ring of one-lane segments with finite bays deadlocked in 3.9 % of its encounters, and the landing closure four times in 10,000 at a crossing on the zone boundary. The exhaustive search of 47,085 reachable states found no unsafe or stuck state and caught three removed rules. At today's fleet the rules added 0.1 % to delay per delivery and no deadlocks, and 7.9 % at four times. Referring every conflict to Earth raised delay per delivery 5.1-fold. On Mars the rules behaved as on the Moon, and referral to Earth cost about one-sixth as much. A base whose robots settle conflicts themselves pays almost nothing for safety; one that keeps the decision on Earth pays in every blackout. They are a proposal for the agreements the Artemis Accords leave open.

Keywords: lunar surface traffic, multi-robot coordination, runtime assurance, formal verification, rules of the road, Mars surface operations

1. Introduction

Within a few years the roads of a lunar south-pole base will carry rovers, cargo vehicles and small robots that belong to different companies. NASA's Lunar Terrain Vehicle task orders of May 2026 went to two companies for two different rovers, cargo rovers are company designs, and small rovers and hoppers come from other firms again [54]. Each will be driven or supervised from its owner's control room on Earth. Where their routes cross, where a road narrows to one lane between a crater rim and a slope, where a lander is about to come down, and where crew are working on foot, somebody has to decide who waits. Today nobody has written down how.

The policy layer exists and is deliberately thin. The Artemis Accords ask signatories to declare safety zones around their activities, to notify one another and to coordinate before entering, and they say that zones are temporary and not exclusive [1]. NASA's own policy analysis adds three facts that shape any traffic rule: transit corridors near the south pole can be as narrow as 20–40 m in places; nobody agrees on how far to stand from a landing, so a simple standard distance is suggested; and once three or more actors operate, some central coordination “may become necessary” [2]. A conference paper of 2025 argues for lunar rules of the road modelled on the maritime rules, without numbers [3]. None of these documents says what a rover should do when it meets another owner's rover on a one-lane road, how close a robot may pass to a crew member, or what happens to a vehicle that loses contact.

Earth has rule sets for machines that share space, and each solves part of the problem. Warehouse fleets move hundreds of robots over a grid with a central planner and a reservation table, and the execution layer that makes those plans safe on real robots is well understood [4, 5, 6]. Autonomous intersection management gives cars time–space reservations at a junction and shows that a vehicle which never assumes an unconfirmed reservation cannot collide because of a lost message [7]. Drone traffic management lets operators that share no controller file their intentions with federated services, check them against one another, and monitor conformance [8, 9, 10]. Runtime assurance puts a small verified safety program between an untrusted planner and the vehicle [11, 12, 13], and responsibility-sensitive safety gives closed-form safe distances for road cars [14]. What none of them covers together is a surface with long one-lane roads, vehicles of very different speeds and braking, landing zones that close for an hour, crew on foot, radio that fails behind terrain, and owners who do not share a planner. Lunar braking is the physical reason the car rules do not transfer: friction braking scales with gravity, so a rover at 15 km/h needs about six times the stopping distance it would need on Earth, and more downhill [15, 16].

We also wanted to know what the rules cost, because a rule that stops everything is trivially safe. The safety literature measures the cost of a safety layer as task performance with and without it on the same scenarios, as how often the layer overrides the planner, and as deadlocks it introduces [13, 6]. For a lunar base the natural measure is cargo delay, which our earlier papers computed for the same base, vehicles and demand [40, 41, 42]. Our question in this paper is therefore in three parts. Can a small, shared rule set keep robots of several owners safe on real lunar terrain against any planner? What does it cost in delay at today's fleet and at larger ones? And does it still hold on Mars, where gravity is 2.3 times the Moon's, Earth is minutes away and contact is lost for two weeks in every 26 months?

We built the rules into the simulation of [40, 41, 42] at three levels. A discrete model with two or three vehicles, one landing and one crew member is searched exhaustively against a planner that may propose anything; removing a rule must produce a counterexample, or the check proves nothing. An encounter simulator runs continuous kinematics at 10 Hz through the same safety program on twelve types of encounter, from following on a 20° slope to crew emerging from behind a boulder and a planner that proposes random or adversarial moves, and bounds the violation rate per type by the rule of three. The logistics model of our earlier papers runs the base for a year with the road layer on and off, with the same seeds, under two coordination modes: one shared reservation table over the robots' radio mesh, and owners who each see only their own reservations and refer conflicts to Earth. The road graph comes from the base's route network on the 5 m terrain, with one caveat we state up front: at the rovers' 20° slope limit the terrain shows no corridor narrower than two vehicles, so the share of single-lane road is an assumed factor that we vary from none to 40 %, marking the narrowest places first (Section 3.1). The hypotheses, comparisons and statistics were written down before the main runs [43].

The results are these. The exhaustive search of the discrete model found no unsafe state, no deadlock and no livelock in 47,085 reachable states over four layouts, every state could reach the goal within 41 steps, and each of three removed rules produced a counterexample (Section 5.7). In 274,580 driven encounters of twelve types the safety program allowed no hard violation in eleven types, with a bound on the violation probability of 1.2×10^{-3} or less, jointly at 95 % over the twelve types, under each type's parameter distribution; the exception is a vehicle that stops dead inside a one-lane segment (type 9), where the following vehicle closed below the required gap in 58 of 10,000 encounters and touched it in 7, because the gap rule assumes a leader that brakes rather than stops (Section 5.6). In the logistics model, at today's planned fleet, the rules added 0.01 h to the mean delay per delivery of 15.5 h, 0.1 %, well inside the pre-registered 5 % margin, and the fleet spent 24 h a year waiting at passing bays (Section 5.2). The cost rose with the fleet: 3.7 % at twice the fleet and demand, 7.9 % at four times, and 41.7 % at eight times (Section 5.3). No run at any fleet size deadlocked. The same rules without a

shared table were another matter: owners who referred every conflict to Earth saw delay per delivery rise from 15.5 h to 79.1 h, and their vehicles stood for 1,969 h a year waiting for a decision that could only come while Earth was in view (Section 5.4). A strict reading of the lost-contact rule, in which a robot that cannot reach the base computer files nothing, cost almost as much (Section 5.5). On Mars, with the lunar vehicles on the same roads, the safety program behaved as on the Moon, the dead-stop exception included, but only on the flat: at Mars gravity the lunar brakes hold no downhill steeper than 7.7°, so no sloped encounter could be driven. Coordination that depends on Earth lost about one-sixth of what it loses on the Moon, the opposite of what we pre-registered for Mars (H9, Section 4.1): with Earth in view for half of every sol, as we assumed, a referred conflict waits hours, not the days of the polar base's window gaps (Section 5.8).

Our contributions are:

- **A parameterised rule set for lunar and Martian surface traffic** among vehicles of different owners, with each parameter traced to a source, a derivation or an assumption to be swept: separation from braking physics, single-lane reservations with an exit bay, landing closures, crew zones, lost-contact behaviour, a priority ladder in which owner identity never counts, and conformance.
- **Three kinds of evidence for the same rules.** An exhaustive check of a discrete model against any planner with mutation tests, per-type statistical bounds from simulated encounters, and the delay the rules cost inside a year-long logistics model, all from one implementation of the safety check.
- **A measured price of coordination.** The delay cost of the rules at one, two, four and eight times today's fleet, the congestion peak, and the cost of keeping conflicts on Earth instead of settling them on the Moon, on real terrain with the base's real Earth-link windows.
- **The same rules on Mars.** Gravity, the Earth round trip and the conjunction changed, nothing else, with the lunar vehicles kept on purpose; every Mars encounter and run is paired with its lunar twin.

Several things are outside this paper. Who gets which job, and how robots of different owners share jobs and lift together, is the subject of a companion paper [44]; here every cell uses the same pooled dispatcher, so only the road layer differs between runs. Payments and owners who misreport are the subject of another [45]; here every vehicle obeys the rules and reports its position honestly. The road graph is the base's route network, not a street plan; single-lane road is an assumed share of it. The rules handle vehicles and crew on foot, not landers in flight beyond the closure window, and not the dust a landing raises. The model checker covers a small model exhaustively and the encounter simulator covers twelve families; neither covers real sensors, and we say so in Section 7.

2. Related work

2.1 Fleets on shared roads

Multi-agent path finding gives the planning side of our problem a mature theory. Conflict-based search finds optimal plans for tens of robots [18]; bounded-suboptimal and prioritised planners reach hundreds; priority inheritance with backtracking and the LaCAM family reach thousands of agents on grids in seconds [21, 22]. Prioritised planning writes each robot's plan into a space–time reservation table that later robots must avoid [4, 52]. It is incomplete in general, but complete on “well-formed” layouts in which every robot can wait somewhere that blocks nobody [19, 20], and the priority-inheritance guarantee needs every edge to lie on a cycle, which a dead-end spur or a one-lane corridor between two bays violates [21]. In lifelong operation, where tasks keep arriving, throughput rises with fleet size up to a congestion peak and then falls [6, 17], and planners

with short horizons can deadlock [6]. Two results from this literature matter most here. A plan is safe on real robots only if an execution layer enforces its ordering of who passes first; Hönig and colleagues' action dependency graph does that with light communication and arbitrary vehicle dynamics, and in a warehouse simulation its throughput was 2.5 times that of synchronous execution [5]. And the remedy for narrow roads is structural: a vehicle enters a one-lane segment only when its exit is guaranteed, which breaks the circular wait that every deadlock needs [23]. Our rule R3 is that remedy; Section 3.2 states it and Section 5.3 reports what it cost.

Warehouse fleets are the largest deployed case, with a central job manager allocating tasks and space to hundreds of drive units [24]. Autonomous intersection management is the closest to a rule for conflicting owners: a vehicle requests a time–space reservation from an intersection manager, which accepts it only if no tile is double-booked [7]. Dresner and Stone drew two design rules we adopt. The system must be safe under total communication failure, and a vehicle never assumes a reservation it has not had confirmed, so a lost message can only add delay. They also found that a coarse reservation (the whole intersection as one tile) deadlocked in heavy traffic, and that time buffers proportional to speed beat static ones [7]. All of these systems assume one owner or cooperative vehicles, dense grids, no landing zones, no crew on foot, and no proof over an unknown planner.

2.2 Traffic among operators who share no controller

Drone traffic management is the only deployed template for vehicles of many operators sharing space without a single controller. In NASA's concept every operation is filed in advance as a four-dimensional intent, service suppliers that serve different operators check each new intent against all others with first-come-first-served precedence and priority for public safety, and a vehicle that strays from its intent is broadcast to everyone nearby [8]. The European regulation makes the deconfliction rule explicit: authorised operations must be “free of intersection in space and time” [10]. NASA's field tests of the federated exchange measured an average message latency of 189 ms, with 95 % of exchanges within 531 ms but a 95th percentile of 3.9 s for negotiation messages, and attributed operational impact to downed servers and misimplemented specifications rather than to latency [9]. The interoperability standard adds conformance monitoring and constraint management [25]. What transfers to the Moon is the structure: filed intents, federated checking, a priority ladder, published constraints such as a closed landing zone, and conformance monitoring. What does not transfer is the internet; our robots talk over a terrain-limited radio mesh that fails behind ridges, and the Earth link is seconds away at best and absent for days [42].

2.3 Safety layers that override a planner

Every mainstream safety-layer design shares one pattern: an unverified controller proposes, and a small verified component passes the proposal, changes it minimally or switches to a verified backup, so that the system stays in a set of states from which safety can always be kept. Simplex [11, 53] and the aircraft standard built on it [12] give the switching form; control barrier functions give the minimal-change form as a quadratic program, with a multi-robot version run on real robots [26, 27]; shields give the discrete, temporal-logic form, correcting an action only when it would violate the specification [28]; reachability computes the safe set itself [48]. Responsibility-sensitive safety gives closed-form safe distances for cars from response time and bounded accelerations, with a rule that assigns blame to the vehicle that broke its proper response [14]. Verified ground-robot navigation shows which safety notion is achievable with moving obstacles: passive safety, in which the robot is stopped if a collision happens, unless the other agents also obey rules [29]. A tutorial and taxonomy of these filters is given by Hobbs and colleagues [13], and a comparison on spacecraft docking found the implicit

filters, which simulate the backup trajectory online, 11–13 times costlier in computation than explicit switching [30]. Our safety program is a post-posed shield in the Simplex form: it holds the vehicle's reservations and the published constraints, accepts the planner's proposal only if every rule still holds, and otherwise substitutes the safe fallback. Because other vehicles obey the same rules, we can claim more than passive safety on the discrete model, and we claim it only there.

These papers measure the cost of the layer in a small set of ways: task performance with and without it on the same scenarios; how often and how much the layer changes the command; conservatism of the safe envelope; computation time; and deadlocks or freezing the layer introduces [13, 30, 6]. We use the first and the last, in cargo delay and deadlock counts on the same seeds, and report the computation in a companion paper [45].

2.4 Proving rules and bounding what simulation cannot prove

Two kinds of evidence answer different questions. Exhaustive verification proves that the rules, as modelled, never allow a violation and never deadlock, on a discretised model; Amazon's experience with TLA+ shows that such models find bugs that testing misses, with error traces 35 steps long [31, 50], and timed-automata and probabilistic checkers have been applied to intersection protocols [32, 46]. Simulation then measures how the implemented system behaves on continuous dynamics, including what the model abstracted away. Statistical model checking makes that second step a bounded claim: it samples runs of a system too large to search exhaustively and states, with a confidence level, a bound on the probability that a property is violated [47]; our per-type bounds are that procedure with a zero-numerator estimator. The statistics of zero violations are simple and often misread: with no events in n independent trials the one-sided 95 % upper bound on the per-trial rate is $1 - 0.05^{1/n}$, about 3/n [33]; to demonstrate that a fleet is safer than human drivers would take hundreds of millions of failure-free miles [34]; and the bound holds only for the distribution the generator produces, which is why rare encounters are sampled at boosted rates and reweighted [35] and why stratified encounter types are reported separately. Signal temporal logic turns a pass or fail into a signed margin, so a reader can see how close the system came [36]. A credible claim therefore states the property and the model, the reachable states and the counterexamples found during development, the encounter types with their counts and bounds, near misses and minimum margins, the cost of safety on the same seeds, and what is not covered. Section 4 follows that list. One rule from this literature shapes our implementation: the checked code must be the running code, so our model checker, encounter simulator and logistics model import the same safety check.

2.5 The Moon and Mars

Lunar policy gives the hooks and the numbers but not the rules. The Artemis Accords define safety zones as areas where nominal operations or an anomaly “could reasonably cause harmful interference”, sized by accepted engineering principles, temporary, and to be entered only after notification and coordination [1]. NASA's policy analysis states that zones are not exclusion zones, recalls the Apollo heritage boundary of 2 km around landers because 1–10 cm ejecta reached about 1.5 km, reports transit corridors modelled as narrow as 20–40 m, recommends standard standoff distances until plume models improve, and expects central coordination to become necessary with three or more actors [2]. NASA's logistics guidance lists a lander ejecta separation above 1,000 m [38], and crater rims shelter a rover within about 200 m of a lander [37]. The reference vehicles drive at up to 15 km/h on slopes to 20° [16]; the Apollo 16 crew reported stopping from about 10 km/h in roughly twice the distance of the one-g trainer, because braking force scales with gravity [15]. We found no published requirement for the distance an uncrewed vehicle must keep from crew on foot; the closest analogue is the industrial-truck standard, which limits speed to 0.3 m/s where personnel detection is muted [58]. Surface radio is

immature: the only cellular network flown ran for about 25 minutes without completing a call [49], and the three-rover mesh of CADRE has not yet flown [39]; the LunaNet specification leaves the surface-to-surface link to bilateral agreement [51]. A 2025 conference paper raises priority of passage, passing, signalling and crew clearance as regulatory questions and proposes the maritime rules as a model, qualitatively [3]. Mars adds gravity of 3.71 m/s^2, a round trip to Earth of 6–44 minutes and a solar conjunction of about two weeks every 26 months with no contact at all; we found no surface traffic study for either body.

2.6 Our earlier work and the gap

Our first paper built a discrete-event simulation of cargo movement at a south-pole base and a pooled dispatch marketplace [40]. The second added landing precision, pad standoff, a plume keep-out window and rover autonomy on NASA's 5 m terrain of the Connecting Ridge [41]. The third computed the base's Earth-link windows from ephemerides and terrain and priced their gaps [42]. The companion paper of this series moves dispatch from Earth onto the robots over a radio mesh computed from the same terrain and measures what that costs [44]. In all of them vehicles did not interfere with one another on the road: the road layer of this paper is the first time two vehicles in our model have had to decide who waits.

Table 1 sets the closest works against the features our question needs. We did not find a rule set for surface traffic among vehicles of several owners that is checked exhaustively against any planner, tested in simulated encounters with per-type bounds, and priced in logistics delay on real planetary terrain with real Earth-link windows. Each piece is published somewhere; the combination, and the lunar and Martian numbers inside it, are what this paper adds.

Table 1. Features of the closest prior work. “Yes” means the feature is part of the study's model, proof or experiment; “–” means it is absent.

Work	Several owners	One-lane roads	Landing closures, crew	Lost-contact rule	Proof against any planner	Per-type statistical bounds	Cost in task delay	Planetary terrain
Hönig et al. 2019 [5]	–	Yes (grid)	–	–	–	–	Yes (throughput)	–
Li et al. 2021 [6]	–	Yes (grid)	–	–	–	–	Yes (throughput)	–
Dresner and Stone 2008 [7]	Yes (cars)	–	–	Yes	–	–	Yes (delay)	–
NASA UTM, TCL4 [8, 9]	Yes	–	Yes (constraints)	Yes (contingency)	–	–	–	–
Shalev-Shwartz et al. 2017 [14]	Yes (cars)	–	–	–	Yes (closed form)	–	–	–
Alshiekh et al. 2018 [28]	–	–	–	–	Yes (shield)	–	Yes (reward)	–
Mitsch et al. 2017 [29]	–	–	–	–	Yes (hybrid proof)	–	–	–
Zhao et al. 2017 [35]	–	–	–	–	–	Yes	–	–
NASA OTPS 2022 [2]; Kim et al. 2025 [3]	Yes (policy)	Yes (named)	Yes (named)	–	–	–	–	Yes (named)
Our earlier work [40–42, 44]	Yes	–	Yes (plume window)	–	–	–	Yes	Yes
This paper	Yes	Yes	Yes	Yes	Yes (discrete model)	Yes	Yes	Yes, Moon and Mars

3. Model

Three models share one implementation of the rules. The logistics model is the discrete-event simulation of our earlier papers [40, 41, 42], with a road layer added that makes vehicles wait for one another; it prices the rules in cargo delay over a year. The encounter simulator runs two to four vehicles, a landing and a crew member through continuous kinematics on a small piece of road, through the same safety check, and counts violations. The discrete model is a small abstraction of the same situations, searched exhaustively. This section describes the base and its roads, the rules, the safety program, the two coordination modes, and the two smaller models. Provenance follows the convention of [41]: *Measured* values come from a cited measurement or document, *Spec* from a published specification, *Derived* from a computation on such values, and *Assumed* values are our choices, each varied in Section 4.2 or named in Section 7.

3.1 Base, terrain and roads

The base is the one of [41, 42, 44], on the Connecting Ridge between Shackleton and de Gerlache craters, on NASA Goddard's 5 m terrain tile from the Lunar Orbiter Laser Altimeter (Measured). Its sites are the habitat and control centre (BASE), a cargo depot, a power plant, a construction site, an oxygen pilot plant, a science site and a permanently shadowed crater; landers use two pads 1 km from the base. Routes between sites are the driving routes of [41], sampled every 100 m along their length: 2,421 samples on 242 km of distinct road, with 17 junction samples where routes meet. Demand, landings and the fleet are those of the companion paper at its middle load [44]: six vehicle classes of 2028–2032 programmes, from 80 kg crew rovers to a 2,000 kg cargo rover, owned by three companies, busy about 30 % of the cargo vehicles' time (load factor 2.806, Assumed, calibrated in [44]). Larger fleets in Section 4.2 multiply every vehicle and the demand by the same factor.

Which parts of these roads are one lane wide is a question the terrain can only partly answer. We measured the drivable corridor at every sample as the width, across the route direction, over which the slope on the 5 m grid stays below a limit, and called a sample single-lane when that corridor is narrower than two vehicle widths plus two lateral margins: 2 × 2.3 m + 2 × 2.7 m = 10 m (Spec for the width of the reference rover [16]; Assumed for the margin). At the rovers' 20° slope limit no sample is single-lane: the median corridor is 101 m wide and the steepest slope on any route is 17.2°. At a 15° limit 0.4 km of road is single-lane, at 10° 36.5 km (15.1 % of samples), and at 8° 96.5 km. The 5 m grid cannot see boulders, crater rims, shadows or a graded one-lane road, all of which narrow a real corridor, and NASA's own corridor modelling found sections 20–40 m wide [2]. We therefore treat the single-lane share as an assumed scenario factor: a share s marks the s × 2,421 samples with the narrowest corridor at the 10° criterion as single-lane, so the places that go first are the real pinch points of the route network, with s = 0.15 by default (Assumed; the 10° value) and s from 0 to 40 % in Section 4.2. The ranked places, with coordinates, are in the repository. Consecutive single-lane samples along any route form a physical segment; passing bays split segments every 1 km (Assumed; 0.5–2 km in Section 4.2). Chargers and the pads' aprons are reserved slots, two per pad (Assumed).

3.2 Rules of the road

Table 2 states the ten rules, their default parameters and the basis of each. They are written so that a vehicle can check every one of them from what it knows locally: its own state, the reservations it holds, the reservations and intents its neighbours have broadcast, and the published constraints. Four principles run through them. A vehicle never assumes a reservation it has not had confirmed, so a lost message can only add delay [7]. A vehicle enters a one-lane segment only with the whole segment reserved for its traversal and its exit guaranteed, so no vehicle ever waits inside a segment and a circular wait cannot form [19, 23]. Priority follows a fixed ladder in which

crew on foot and landings come first and the identity of the owner never counts, with ties broken by the order in which the base computer queued the requests and then by vehicle number, so no conflict is left open [8]. And the rules are the same for every owner, which is what lets a safety program on one vehicle reason about what the others will do.

Table 2. Rules of the road with their default parameters. Level: L = enforced in the logistics model (Section 3.4), E = in the encounter simulator and the discrete model (Sections 3.5, 3.6). Basis: Sourced = from the cited document, Derived = computed from sourced inputs, Assumed = our choice, varied where Section 4.2 says so.

#	Rule	Default	Level	Basis
R1	**Keep distance, same direction.** Gap to the vehicle ahead at least the responsibility-sensitive safe distance for response time ρ, own maximum acceleration a_{acc}, own guaranteed braking b_{min} (the smaller of the service brake and the friction limit, reduced by $g \sin\theta$ downhill), the leader's maximum braking b_{max}, plus a static margin d_0	ρ = 1 s, a_{acc} = 0.2 m/s^2, b_{min} = 0.5 m/s^2 on the flat, b_{max} = 0.8 m/s^2, d_0 = 5 m: 6.9 m + d_0 at 10 km/h, 12.5 m + d_0 at 15 km/h	E	Formula Sourced [14]; braking $b = \min(b_{service}, \mu g \cos\theta) - g \sin\theta$, the smaller of the 0.5 m/s^2 service brake and the friction limit, less the downhill term, with $b_{service}$ and μ (0.3–0.6) Assumed
R2	**Speed from stopping and sight distance.** Stopping plus reaction distance within the clear sight distance along the road, and within the class speed limit; downhill slopes over 10° capped at the speed that stops in 15 m	Class limits: crew rovers 15 km/h, cargo rovers 4–6 km/h [16]	E	Apollo 16 braking Sourced [15]; table Derived; the limited-visibility rule of [14]
R3	**Head-on on one-lane segments.** A vehicle enters a one-lane segment only with an exclusive reservation for the whole segment and its time window, ending at a bay or junction with a free slot; vehicles in the same direction may follow	Segment: corridor under 10 m (Section 3.1); bays every 1 km	L, E	Well-formed layouts [19, 20]; circular wait [23]; corridors of [2]
R4	**Give way on narrow roads.** Conflicting requests for a segment follow the ladder of R8; within a class, the loaded vehicle, then the request the base computer queued first (its queue order, not the filing clock, so a vehicle further back can never be granted a slot ahead of one nearer it), then the lower vehicle number; earlier commits keep their reservations	No tie left open	L, E	First-come-first-served with priority [8]; maritime narrow-channel rule by analogy [3]
R5	**Landing-zone closure.** No vehicle within R of an active pad from T_{pre} before touchdown to T_{post} after; a vehicle inside at the notice drives to the boundary or to a sheltered spot	R = 1.0 km, T_{pre} = 30 min, T_{post} = 15 min	L, E	Standoff over 1,000 m Sourced [38]; Apollo 2 km boundary and the standard-distance suggestion [2]; times Assumed; the plume window of [41]
R6	**Keep clear of crew.** Never closer than R_{stop} to crew on foot; within R_{slow} drive at v_{crew} and yield; crew always have the right of way	R_{stop} = stopping distance + 10 m, R_{slow} = 50 m, v_{crew} = 1 km/h; EVAs twice a week for 6 h at the habitat	L (slow zone), E	No lunar requirement found; industrial-truck analogue [58]; schedule Assumed
R7	**Lost contact.** A vehicle out of radio contact keeps only the reservations it holds; when the last expires it stops at the next safe point outside segments and zones; others treat its last reserved volume as occupied until it reappears or a hold timeout passes	Hold 10 min; in the logistics model, a job's reservation travels with its assignment (default) or is filed only from a place the base mast can see (strict variant)	L, E	Lost-message rule of [7]; contingency handling of [8]
R8	**Priority ladder.** Crew on foot; lander in descent or ascent; crewed vehicle; vehicle with restricted manoeuvrability (loaded hauler, vehicle on a slope over 15°, vehicle with a fault); uncrewed loaded; uncrewed empty; parked or charging	Owner identity never counts	L (urgent cargo, loaded before empty), E	Maritime hierarchy by analogy [3]; public-safety priority of [8]
R9	**Conformance.** Every vehicle broadcasts its state and filed intent; a vehicle more than e_{conf} from its intent or more than t_{late} behind its reservation loses its forward reservations after a grace period, and neighbours inflate its occupied volume	1 Hz; $e_{conf} = 3\sigma_{nav} + 5$ m = 11 m; t_{late} = 60 s	E	Conformance monitoring of [25, 10]; values Assumed
R10	**Pads, chargers, junctions.** Junctions are reserved as tiles with time buffers; chargers and pad aprons are reserved slots with a maximum dwell	Two apron slots per pad; even tile counts at junctions	L (aprons), E	Tile granularity result of [7]

3.3 The safety program

Each vehicle runs a planner and a safety program. The planner may be anything: the hand-built driving logic of [41], a learned policy, or, in the tests of Section 3.5, a generator of random and adversarial proposals. It proposes the next action as a road segment, a speed and a decision to go or wait. The safety program holds the vehicle's reservations, the broadcasts it has heard and the published constraints, checks the proposal against every rule of Table 2 that applies at its level, and either passes it or substitutes the safe fallback: slow to the safe speed, hold at the current safe point, or stop. It is the post-posed shield of [28] in the switching form of [11, 12]: it never plans, it only refuses. One function implements the check, and the logistics model, the encounter simulator and the discrete model all call it, so what is checked is what runs.

3.4 The road layer in the logistics model

The logistics model moves cargo through the base for a simulated year with the pooled dispatcher of [40], which sees every vehicle and every item. The road layer sits between dispatch and driving. When the dispatcher commits a job, the layer walks the vehicle's planned legs over the route samples and applies, in time order, the rules that act at the resolution of a 100 m sample and a job: the one-lane reservations of R3 with the ladder of R4 and R8, the apron slots of R10, the crew slow zone of R6 around the habitat during a scheduled EVA, and the landing closure of R5, which is the plume window of [41] unchanged. A reservation covers the whole segment for the vehicle's traversal window and ends at a bay or junction, so the exit-bay rule holds by construction, no vehicle waits inside a segment, and because the reservation table is filled one job at a time, a circular wait cannot arise. A conflict is a reservation in the opposite direction that overlaps in time; the later request waits at the bay until the segment is free, and the wait extends every later leg of the job. Following vehicles in the same direction are not separated at this resolution: R1 and R2 act within a segment, below the model's grain, and belong to the encounter simulator.

Two coordination modes run the same rules. In the *ATC* mode there is one shared reservation table, kept by the base computer and reached over the robots' radio mesh of [44]; a reservation is granted in milliseconds when the robot is linked. By default the reservation travels with the job assignment, which reached the robot, so every job files at commit; in the strict variant of R7 a robot files only from its start site or the first route sample the base's mast can see, and a job that cannot file goes to the second mode's procedure. In the *uncoordinated* mode each owner keeps its own table and sees only its own vehicles' reservations. A conflict between vehicles of different owners is detected when they meet: both stop at the conflict time and wait until the owners' operators on Earth settle it by hand, which takes t_{hand} = 2 h (Assumed) and can only begin while the base sees Earth, with the Earth-link windows of [42]. The other vehicle's stop lengthens its busy time; deliveries it had already recorded are not moved, which understates the uncoordinated mode's cost and is listed in Section 7. A third setting, *none*, removes the road layer and the landing closure altogether and is the baseline against which the rules are priced.

3.5 The encounter simulator

The encounter simulator runs two to four vehicles on a small local road geometry with continuous two-dimensional kinematics at 10 Hz, with the class speeds of Table 2 and guaranteed braking $b = \min(b_{service}, \mu g \cos\theta) - g \sin\theta$, the smaller of the 0.5 m/s^2 service brake and the friction limit $\mu g \cos\theta$, less the downhill term $g \sin\theta$, with $b_{service}$ Assumed and friction μ swept over 0.3, 0.45 and 0.6 (Assumed; the regolith braking coefficient is not published, Section 7). Each vehicle's planner is one of three: the nominal driving logic, a planner that proposes a random action at every step, and an adversarial planner that proposes the action closest to

a violation. Every proposal passes through the safety program of Section 3.3. Twelve encounter types (Table 3) are generated with randomised parameters, one encounter per seed; the closest approach between steps is computed exactly from the kinematics, not sampled; rare types are generated at boosted rates and reweighted, with the weights reported. Types 1 and 7 run 100,000 encounters each and the other ten 10,000 (change 6 of [43], a cost of compute on the shared machine that weakens those bounds tenfold); type-1 draws whose slope and friction cannot hold the vehicle (25 % of them) are not driven and are counted apart. Each type's bound is the 95 % upper limit on the probability of a hard violation under that type's parameter distribution: with no violation $1 - 0.05^{1/n_{\text{eff}}}$ and, beside it, the Bonferroni limit $1 - (0.05/12)^{1/n_{\text{eff}}}$ over the twelve types; with violations the Clopper–Pearson upper limit on the unweighted count at the same levels, with the limit on the weighted violation share at the effective n reported beside it. Types 6, 7 and 9 are importance-sampled (the inside vehicle's exit distance, the crew's emergence time and the dead-stop time) with weights p/q and an effective n of $(\Sigma w)^2/\Sigma w^2$; type 12 fuzzes draws of types 1, 2, 4 and 7 and so inherits type 1's undriven slopes and type 7's weights. A near miss is a same-lane gap below twice d_{min}, a time to collision below 10 s, a crew clearance below twice the stopping radius, or a zone exit margin below 300 s; the minimum margin is the smallest clearance to another vehicle or crew member. Every encounter receives exactly one of six outcomes by precedence (not driven, violation, blocked by a fault, deadlock, finished, incomplete); deadlock is judged only where no random fuzzer runs, and an emergency-brake event (no candidate satisfies every rule and the shield applies full braking) is a rate, not an outcome. The first main run was stopped when two simulator defects showed as violations, the vehicle body swinging off the path at corners and type-6 crossings built without a junction tile; both were fixed, the stored encounters regenerated and all twelve types rerun at the same n (change 8 of [43]); the stopped run's counts are reported in Section 5.6 and its files archived. During the rerun the type-6 cell ran three times: its first pass showed a third defect, start states inside a vehicle's own stopping distance of a crossing tile, and a same-lane test in the gap metric wide enough to count two vehicles leaving a shallow crossing on different roads; its second pass exposed a stall classifier that counted a vehicle correctly held at a closed zone as waiting for a grant. The builder and the classifier were fixed, the metric's same-lane test narrowed to the same road and direction (the shield's own leader test is unchanged), and types 3, 5 and 6 rerun at the same n so that every count is under the one final metric and classifier (change 9 of [43]). Twenty stored encounters must reproduce exactly on every build.

Table 3. Encounter types of the simulator and the discrete model. Each type is generated with randomised positions, speeds, slopes and timings within the stated ranges; the number of encounters per type and the resulting bounds are in Section 5.6.

#	Type	What it tests
1	Following on the flat and on 10°, 15° and 20° downhill; the leader brakes at maximum	R1, R2
2	Head-on on a one-lane segment, with and without a bay within reach, with a third vehicle queued behind each	R3, R4
3	Overtaking on a two-lane stretch ending before a one-lane segment	R1, R3
4	Crossing and merging at a junction; four-way contention for one junction	R10
5	Cyclic wait: three or four vehicles each holding a segment the next needs	R3 (deadlock)
6	Landing announced with a vehicle inside R, with a route filed through R; landing delayed and waved off	R5
7	Crew crossing a road; crew emerging from behind a boulder; crew walking towards a stopped robot	R6
8	Dropout mid-negotiation (one side receives the grant, the other does not); 10–50 % message loss; a partitioned mesh	R7
9	Nonconforming vehicle: late from a slip, off-route from navigation drift, stopped dead inside a segment	R9
10	Mixed owners and classes on one segment; one owner's planner always aggressive	R4, R8
11	Charger and pad contention at two, four and eight times the fleet	R10
12	Planner fuzzing: random and adversarial proposals at every step	All

3.6 The discrete model and its checker

The discrete model is a road graph of 12 nodes with typed edges (two-lane, one-lane) and nodes (junction, bay, parking, pad boundary), including one one-lane segment of four cells with a bay at each end and one junction; two or three vehicles, each with a cell, a heading, a mode (drive, wait, yield, stopped) and its held reservations; one landing with phases open, notice, closed and reopen; one crew member moving on a few cells; and a link state per vehicle. Time is one cell transit of the slowest class. Transitions interleave vehicle moves, crew moves, landing phases and link drops, all nondeterministic, and the planner is fully nondeterministic, so whatever the search proves holds for any planner. Each vehicle move goes through the same safety check as the simulators. The checker is an explicit-state search in the style of [50, 32]: states packed into integers, vehicles of the same class interchanged by sorting, a hard cap of three million states. It checks the safety invariants on every reachable state (no two vehicles in the same or adjacent cells on a segment, no vehicle in a closed zone, none in a crew cell or next to it), deadlock freedom (every reachable non-goal state has an enabled move that is not "all wait"), bounded liveness (from every reachable state a state where all vehicles have reached their goals is reachable under fair scheduling), and livelock (strongly connected components with no exit). A checker that never fails proves nothing, so we remove one rule at a time, R3's exit reservation, R5's entry check and R7's hold, and require a counterexample each time.

3.7 Mars

The Mars variant changes five things and nothing else. Gravity is $g = 3.71$ m/s^2 in the braking and slope terms of R1 and R2. The Earth round trip, 6, 20 and 44 minutes, is added to the two hours of staff time for every conflict referred to Earth, in place of 2.6 s. The Earth windows, which on the Moon are the computed terrain-masked windows of [42], are on Mars assumed to be the first half of every sol (Assumed, Section 7), or the whole sol outside the conjunction and the storms when one relay in Mars-synchronous (areostationary) orbit is added as a variant. A solar-conjunction blackout of about 14 days comes on Mars once in about 26 months, and the simulated year holds one. And two five-day dust storms are modelled as loss of the Earth link only (Assumed), with the robots' radio mesh unchanged. The road geometry is the Connecting Ridge network with Mars parameters, as the controlled case; a second geometry on a Mars site was planned and not run, because the terrain reader is written for the lunar polar tiles and a Jezero base would need its own reader, sites and route network (Section 7).

3.8 Checks

The engine's existing checks are those of [40, 41, 42, 44]. The road layer adds its own: a single vehicle on the network sees zero added delay from the road layer; a run with the single-lane share at zero and no landings reproduces the baseline's deliveries; reservations never overlap in the shared table; a run is bit-identical when repeated with the same seed; and every row of every run file carries the cell's settings, so a result cannot be matched to the wrong cell. In addition, 86 checks of the engine's phase-5 verification suite (`lunarsim/vv5.py`, results in `results/vv_phase5.json` and on the live site's verification page) bear on this paper: 86 verifications that the code does what the model says, by Sargent's technique analytic (20), traces (18), internal validity (13), comparison to other models (9), degenerate tests (9), fault injection (mutation testing) (8), extreme condition (4), metamorphic testing (4), event validity (1). Deterministic identities are held to 10^{-9} relative or exact; stochastic checks pass when the error is within a tolerance set in advance, and a 99 % batch-means interval that misses the reference is recorded as a bias flag, not a failure; the checks ran at commit `5c5b8fd`. Their outcomes are in Section 5.9.

4. Experiment design

4.1 Questions and hypotheses

The pre-registration [43] states three hypotheses for this paper. **H7 (safety).** Zero violations in each of the twelve encounter types at 10^5 encounters per type, with the bound $1 - 0.05^{1/n}$ reported per type, and a counterexample from the checker for each rule removed. Change 6 of [43], made after the development runs of the simulator and before any main E44 row, cut ten of the twelve types to 10^4 encounters for compute and kept types 1 and 7 at 10^5 (Section 3.5); the bound per type is unchanged and is weaker by a factor of ten at the smaller sample. **H8 (cost of safety).** At today's fleet the rules add less than 5 % to the delay per delivery; at four to eight times the fleet the cost rises. Only the first clause is confirmatory; the second is reported. **H9 (Mars, exploratory).** The same rules keep zero violations under Mars gravity and braking; the delay cost of the rules is similar to the Moon's; and coordination that depends on Earth loses more on Mars than on the Moon because of the round trips and the conjunction. H7 and H8 were written before any run; H9 was added with the Mars part on 27 October 2026, before any run, when the three papers were regrouped (change 1 of [43]).

4.2 Experiments

E43, the price of the rules. Mode {none, ATC, uncoordinated} by fleet density {1, 2, 4, 8} times the planned fleet and its demand, with the pooled dispatcher in every cell. The none and ATC modes never wait for Earth, so their runs do not depend on the start year and run for 2028 only; the uncoordinated mode runs 2028 and 2038, a good and a poor year from the window set of [42] rather than its extremes: the base sees Earth in 61.5 % of the hours of 2028 and 54.7 % of the hours of 2038. Seeds 1–30 at one and two times the fleet over a full year; seeds 1–10 at four times over 90 days and at eight times over 60 days. Those horizons and seed counts were fixed in the design before the dispatch fix of Section 4.6 and kept afterwards so that the rerun matched the archived rows; Section 7 lists the shorter horizons as a limitation. Sweeps at one times the fleet, 2028, 30 seeds, in both coordination modes: single-lane share 0, 0.05 and 0.40 against the default 0.15; bay spacing 0.5 and 2 km against 1 km; and the strict reading of R7 in the ATC mode. 650 runs in 27 cells. **E44, encounters.** 300,000 encounters in the twelve types of Table 3 (100,000 in types 1 and 7, 10,000 in each other), with the three planners of Section 3.5, importance weights on types 6, 7 and 9, and a 95 % per-type bound with a Bonferroni correction over the types. **E45, the exhaustive check.** Four layouts of the discrete model of Section 3.6: a pad visit with one oncoming vehicle (landing, crew and radio links); three vehicles on the line, two of them interchangeable (crew, no landing); a pad visit with two oncoming vehicles (landing and crew); a pad visit with two oncoming vehicles and radio links. Then the three mutations, R3 without the exit-slot check, R5 without the entry check, R7 without the hold, each required to produce a counterexample on the layout where the rule acts. **E47, Mars.** E44's twelve types at $g = 3.71$ m/s^2 with the lunar vehicles unchanged, 2,000 encounters per type and 12,000 for types 1 and 7, generated from E44's seeds so that each Mars encounter is paired with the lunar encounter of the same type, batch and index; and E43's ATC and uncoordinated cells at today's fleet, seeds 1–30 (the 2028 label marks the paired lunar runs; the Mars link uses no start year), with a Mars Earth link: round trips of 6, 20 and 44 minutes added to the two hours of staff time per referred conflict, Earth in view for the first 50 % of each sol (Assumed) or, as a variant, continuously through one areostationary relay, a 14-day conjunction from day 200 and two five-day dust storms, both as loss of the Earth link only. 44,000 encounters and 210 runs in seven cells.

4.3 Seeds and common random numbers

Every cell of E43 uses the same seeds, so a seed's demand, landings and faults are identical across modes and the difference between two modes on one seed is the effect of the road layer alone. The engine's phase-5 check D10 (Section 3.8) confirms the pairing on the companion paper's regimes: the demand stream, landings and manifests, was identical across 880 regime pairs on the same seed and start year, while the per-job intervention and fault redraws, shared in draw order, diverge once the allocations differ; E43's modes draw from the same streams, so a pair shares its year and diverges in those draws after the first conflict. Pairs are formed on the seed; because the none and ATC runs do not depend on the start year, an uncoordinated run of 2038 pairs with the none and ATC runs of the same seed. Encounters use one seed per encounter, so each is an independent trial. No run is excluded after the fact; a run that crashes is fixed and repeated with the same seed, and the crash is reported.

4.4 Outcomes

The primary outcome of E43 is the mean delay per delivered item, in hours from the item's readiness to its delivery, the measure of [40]. Secondary outcomes are the 90th percentile of that delay, the mass-weighted delay, the censored mean wait of urgent items of [44], items delivered, operator hours, and the road layer's own counts: hours waited at bays, conflicts, conflicts referred to Earth and the vehicle-hours spent waiting for Earth, hours other vehicles stood because of a stopped vehicle, deadlocks, and bay wait per owner. The primary outcome of E44 is the count of violations per type, with near misses and the minimum separation margin; of E45, the reachable state count and the properties proved.

4.5 Statistics

Differences between modes are paired means over seeds with a percentile bootstrap 95 % confidence interval of 20,000 resamples [55]; the ratio of a difference to the baseline's mean carries its own bootstrap interval. H8's first clause is supported if the upper end of the two-sided 95 % bootstrap interval of the ratio (ATC minus none, over the mean of none) at one times the fleet lies below 0.05. That is this paper's reading of the registered “add less than 5 %” [43]; the one-sided 95 % bound gives the same verdict, which the build script checks. It is a margin test in the spirit of equivalence testing [56]: as in Schuirmann's two one-sided tests, the claim is that the cost lies within a stated margin, not that it is zero, so the end of the interval nearest the margin decides. The density ladder, the uncoordinated mode and the sweeps are reported with intervals and not tested. Deadlocks are counted as a check of the implementation: the road layer cannot deadlock by construction (Section 3.4), so a non-zero count would be a defect in the code, not a property of the rules, and the count says nothing about finite bays, which only the encounter simulator and the checker have. For E44 the per-type bound with zero violations is $1 - 0.05^{1/n}$ [33], and the joint statement over the twelve types uses $\alpha = 0.05/12$ per type, which raises each bound from about 3/n to about 5.5/n; with any violation the Clopper–Pearson interval replaces it [57]. All intervals are descriptive of the simulated population of seeds, not of a lunar fleet.

4.6 Choices made after looking at results

Three choices were made after looking at data, all before the runs they affect, and all are recorded in [43] and in the design file of E43. First, the single-lane share became an assumed scenario factor after the corridor analysis of Section 3.1 showed no one-lane road at the rovers' slope limit; the default and the sweep were fixed before the first E43 run. Second, the horizon of the eight-times cells: it was planned at 60 days on ten seeds, cut to 30 days when the first 60-day run took 310 s, over the per-run limit of Section 4.7, abandoned when a 30-day run went past 470 s, and restored to 60 days once the defect below was fixed; no 30-day row was ever used, and every

eight-times row in this paper is a 60-day row. Third, a defect in the engine's event handling, found while timing those cells: the dispatcher's consolidation hold re-posted a wake-up for every held lane on every call, so the event queue grew geometrically with fleet size (the cause of the 470 s run above) and, because one dispatch call does not reach a fixed point, the duplicates also changed assignments. The fix, pre-registered as change 5 of [43] before any rerun row, posts one wake-up per lane and lot and repeats dispatch at the same instant until nothing changes. Every cell of E43 was then rerun; the 610 earlier rows are archived and reproducible with the fix off. Against the 610 archived rows with the same cell and seed, the primary outcome moved by −0.13 h on average and by 3.8 h in absolute value, with a largest single-run change of 43 h (in the uncoordinated mode, whose hand-resolution waits depend on the order of events); 69 % of the rows moved by more than 0.1 h. The mean run time fell from 14 to 8 s and the longest run from 394 to 32 s. Table 4 and Table 5 hold only rerun rows; the before/after table per cell is results/E43_fixpoint_before_after.csv.

4.7 Compute

Runs are single-process on one laptop shared with other work, in chunks of at most nine minutes with a per-run limit of five minutes. A year at one times the fleet took 5 s per run, at two times 12 s; the 90-day runs at four times took 8 s; the 60-day runs at eight times took 14 s. E43 took about 1.5 h of processor time. E44's final run took 9.4 h of processor time in resumable batches of 250 to 4,000 encounters, type 11 the longest at 2.7 h, with a process peak of 136 MB. E45's four layouts and three mutations took 50 s in all, the largest layout 37 s and 29 MB.

5. Results

Section 5.1 describes what the road layer did on the base's roads at today's fleet. Sections 5.2 to 5.5 give E43: the primary outcome at today's fleet (H8), the larger fleets, the uncoordinated mode, and the sweeps. Sections 5.6 and 5.7 give the encounter simulator (H7) and the exhaustive check, and Section 5.8 the Mars part (H9). Section 5.9 lists what did not work. Table 4 holds every E43 cell mean and Table 5 every paired difference; Tables 7 and 8 hold the Mars part; the text describes them in that order. Intervals are 95 % bootstrap intervals over seeds unless stated.

5.1 The road layer on the base's roads

At the default single-lane share, 36.3 km of the 242 km of road is one lane wide, in segments split by bays every kilometre. Over a simulated year at today's fleet, vehicles in the ATC mode made 1,843 traversals of those segments, met 73 conflicts, and waited 24 h at bays in all, across the fleet. They waited 0.5 h for an apron slot and drove 10.0 h at crew speed through the habitat zone during EVAs. Landings closed a pad 29 times a year, as in [41]. The bay wait fell on the owners unequally: 19.1 h for owner A, 4.9 h for owner B and 0.4 h for owner C. No run in any cell of E43 recorded a deadlock (0 in 650 runs).

Table 4. Cell means of E43. Delay is the mean delay per delivered item (h) and its 90th percentile; bay wait, conflicts, conflicts referred to Earth and the hours waited for Earth are fleet totals per run; deadlocks are the total over the cell's runs. Horizon in days.

Cell	Start year	Runs	Horizon	Delay (h)	p90 (h)	Items	Bay wait (h)	Conflicts	To Earth	Earth wait (h)	Deadlocks
atc x1	2028	30	365	15.5	34.5	2,927	24.1	73	0.0	0	0
atc x1 bay 0.5	2028	30	365	15.6	34.5	2,927	12.8	107	0.0	0	0
atc x1 bay 2	2028	30	365	15.5	34.5	2,927	24.1	73	0.0	0	0
atc x1 r7	2028	30	365	64.7	196.5	2,928	4.2	16	37.5	1,551	0
atc x1 share 0	2028	30	365	15.5	34.2	2,933	0.0	0	0.0	0	0
atc x1 share 0.05	2028	30	365	15.6	34.5	2,928	24.0	73	0.0	0	0

Cell	Start year	Runs	Horizon	Delay (h)	p90 (h)	Items	Bay wait (h)	Conflicts	To Earth	Earth wait (h)	Deadlocks
atc x1 share 0.4	2028	30	365	15.8	35.4	2,933	42.5	105	0.0	0	0
none x1	2028	30	365	15.5	34.4	2,929	–	–	–	–	–
uncoordinated x1	2028	30	365	79.1	223.7	2,928	0.0	0	52.9	1,969	0
uncoordinated x1	2038	30	365	98.5	254.8	2,915	0.1	0	44.6	2,527	0
uncoordinated x1 bay 0.5	2028	30	365	78.5	223.4	2,925	0.1	0	50.7	1,959	0
uncoordinated x1 bay 2	2028	30	365	79.1	223.7	2,928	0.0	0	52.9	1,969	0
uncoordinated x1 share 0	2028	30	365	15.6	34.4	2,934	0.0	0	1.2	21	0
uncoordinated x1 share 0.05	2028	30	365	79.2	224.5	2,927	0.0	0	53.5	1,968	0
uncoordinated x1 share 0.4	2028	30	365	95.2	250.8	2,917	0.4	2	78.5	2,375	0
atc x2	2028	30	365	8.3	14.5	5,882	287.5	847	0.0	0	0
none x2	2028	30	365	8.0	14.0	5,885	–	–	–	–	–
uncoordinated x2	2028	30	365	64.8	190.2	5,875	64.0	217	238.2	5,020	0
uncoordinated x2	2038	30	365	82.2	228.1	5,857	55.5	186	190.4	6,044	0
atc x4	2028	10	90	5.4	8.0	2,902	384.4	999	0.0	0	0
none x4	2028	10	90	5.0	7.5	2,900	–	–	–	–	–
uncoordinated x4	2028	10	90	63.0	171.3	2,849	92.2	265	206.6	2,793	0
uncoordinated x4	2038	10	90	64.2	169.1	2,888	100.9	254	170.1	3,660	0
atc x8	2028	10	60	5.3	8.5	3,873	1,456.5	2,933	0.0	0	0
none x8	2028	10	60	3.7	5.0	3,870	–	–	–	–	–
uncoordinated x8	2028	10	60	75.6	200.1	3,428	365.9	824	448.3	4,455	0
uncoordinated x8	2038	10	60	67.6	165.0	3,805	427.0	821	361.5	4,680	0

5.2 The price of the rules at today's fleet (H8)

With the road layer off, the mean delay per delivered item at today's fleet was 15.5 h. With the rules on and one shared reservation table it was 15.5 h: a paired difference of 0.01 h [−0.17, 0.22 h], 0.1 % of the baseline, with the upper end of the two-sided 95 % interval of that ratio at 1.4 %. H8's confirmatory clause is supported: the upper bound is below the pre-registered 5 % margin. The 90th percentile of delay moved by +0.07 h [−0.68, 0.89 h], items delivered by −1.8 [−8.3, 4.8] of 2,929, and the censored mean wait of urgent items by −0.41 h [−1.74, 0.98 h]. Operator hours changed by +7.4 h [−2.5, 18.4 h].

Table 5. Paired differences in the mean delay per delivered item (h), 95 % bootstrap intervals over seeds, the difference as a share of the baseline's mean, and the upper end of the two-sided 95 % interval of that share. The H8 rule applies to the x1 row only.

Comparison	n	Difference (h)	95 % CI	Share of baseline	Upper bound
H8: atc - none, x1	30	+0.01	[−0.17, 0.22]	+0.1 %	+1.4 %
uncoordinated - none, x1, 2028	30	+63.53	[58.36, 68.70]	+409.0 %	+446.3 %
uncoordinated - atc, x1, 2028	30	+63.52	[58.40, 68.86]	+408.6 %	+448.6 %
uncoordinated - none, x1, 2038	30	+82.95	[75.98, 89.69]	+534.0 %	+581.1 %
uncoordinated - atc, x1, 2038	30	+82.93	[76.00, 89.75]	+533.4 %	+580.9 %
H8: atc - none, x2	30	+0.29	[0.21, 0.39]	+3.7 %	+4.9 %
uncoordinated - none, x2, 2028	30	+56.81	[52.59, 61.19]	+711.1 %	+769.7 %
uncoordinated - atc, x2, 2028	30	+56.51	[52.35, 60.92]	+682.3 %	+739.0 %
uncoordinated - none, x2, 2038	30	+74.24	[69.51, 79.01]	+929.4 %	+990.4 %
uncoordinated - atc, x2, 2038	30	+73.94	[69.24, 78.74]	+892.7 %	+952.6 %
H8: atc - none, x4	10	+0.40	[0.26, 0.50]	+7.9 %	+10.2 %
uncoordinated - none, x4, 2028	10	+57.99	[47.35, 68.11]	+1,157.4 %	+1,381.0 %
uncoordinated - atc, x4, 2028	10	+57.60	[47.10, 67.58]	+1,065.0 %	+1,262.1 %

Comparison	n	Difference (h)	95 % CI	Share of baseline	Upper bound
uncoordinated - none, x4, 2038	10	+59.16	[49.28, 68.08]	+1,180.6 %	+1,357.1 %
uncoordinated - atc, x4, 2038	10	+58.76	[49.05, 67.87]	+1,086.5 %	+1,269.5 %
H8: atc - none, x8	10	+1.56	[1.17, 1.92]	+41.7 %	+53.9 %
uncoordinated - none, x8, 2028	10	+71.87	[61.56, 81.96]	+1,924.0 %	+2,373.5 %
uncoordinated - atc, x8, 2028	10	+70.31	[60.16, 80.42]	+1,328.2 %	+1,597.6 %
uncoordinated - none, x8, 2038	10	+63.87	[52.97, 77.47]	+1,709.9 %	+2,201.6 %
uncoordinated - atc, x8, 2038	10	+62.31	[51.57, 76.06]	+1,177.2 %	+1,470.4 %
atc x1 bay 0.5 - none x1	30	+0.08	[−0.18, 0.31]	+0.5 %	+2.0 %
atc x1 bay 0.5 - atc x1	30	+0.06	[−0.16, 0.27]	+0.4 %	+1.8 %
atc x1 bay 2 - none x1	30	+0.01	[−0.17, 0.22]	+0.1 %	+1.4 %
atc x1 bay 2 - atc x1	30	0.00	[0.00, 0.00]	0.0 %	0.0 %
atc x1 r7 - none x1	30	+49.16	[43.84, 54.69]	+316.5 %	+352.2 %
atc x1 r7 - atc x1	30	+49.14	[43.77, 54.83]	+316.1 %	+354.1 %
atc x1 share 0 - none x1	30	−0.03	[−0.25, 0.16]	−0.2 %	+1.1 %
atc x1 share 0 - atc x1	30	−0.05	[−0.28, 0.17]	−0.3 %	+1.1 %
atc x1 share 0.05 - none x1	30	+0.03	[−0.16, 0.23]	+0.2 %	+1.5 %
atc x1 share 0.05 - atc x1	30	+0.01	[0.00, 0.04]	+0.1 %	+0.3 %
atc x1 share 0.4 - none x1	30	+0.28	[−0.03, 0.58]	+1.8 %	+3.7 %
atc x1 share 0.4 - atc x1	30	+0.26	[−0.01, 0.55]	+1.7 %	+3.6 %
uncoordinated x1 bay 0.5 - none x1	30	+63.02	[58.15, 68.01]	+405.7 %	+438.6 %
uncoordinated x1 bay 2 - none x1	30	+63.53	[58.54, 68.70]	+409.0 %	+446.5 %
uncoordinated x1 share 0 - none x1	30	+0.06	[−0.19, 0.31]	+0.4 %	+2.0 %
uncoordinated x1 share 0.05 - none x1	30	+63.70	[58.65, 68.83]	+410.1 %	+446.8 %
uncoordinated x1 share 0.4 - none x1	30	+79.70	[74.85, 84.67]	+513.1 %	+547.0 %

5.3 Larger fleets

Fleet and demand were multiplied together, so a larger fleet is also a busier road network. At two times the fleet the rules added 0.29 h [0.21, 0.39 h] to a baseline delay of 8.0 h, 3.7 % (upper bound 4.9 %), with 847 conflicts and 288 h of bay wait a year. At four times, over 90 days, the rules added 0.40 h [0.26, 0.50 h] to a baseline of 5.0 h, 7.9 % (upper bound 10.2 %), with 999 conflicts and 384 h of bay wait in the period. At eight times, over 60 days, the rules added 1.56 h [1.17, 1.92 h], 41.7 % (upper bound 53.9 %). The baseline delay itself fell as the fleet grew. Figure 1 plots the delay per delivery against the fleet multiple for the three modes.

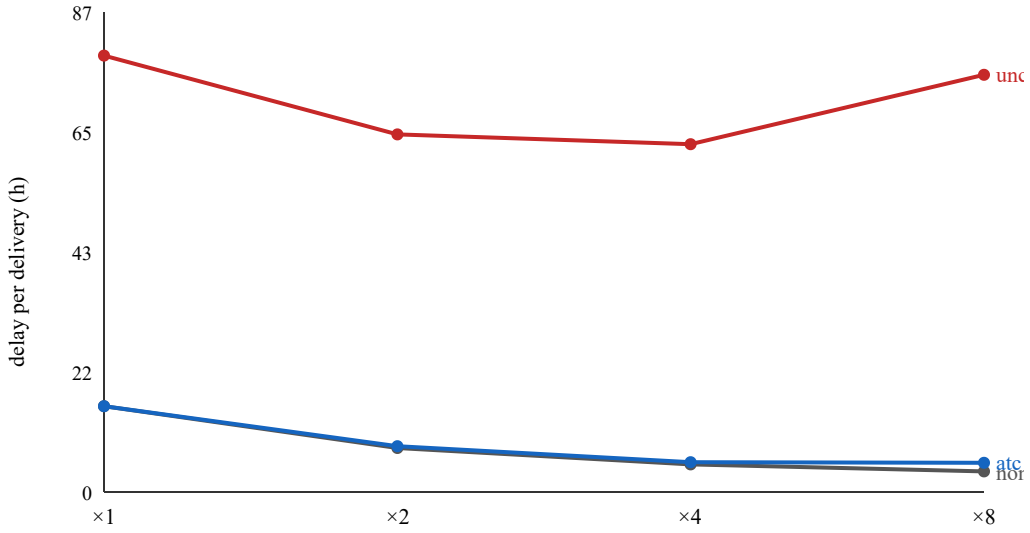


Figure 1. Mean delay per delivered item against the fleet multiple, with the road layer off (none), with one shared reservation table (atc) and with each owner keeping its own table and conflicts settled from Earth (uncoordinated, start year 2028). Cell means over seeds; the four- and eight-times cells run shorter horizons on ten seeds (Table 4).

5.4 Owners who keep their own tables

In the uncoordinated mode, with the same rules but each owner seeing only its own reservations and every cross-owner conflict settled from Earth, the mean delay per delivery at today's fleet was 79.1 h in the 2028 start year and 98.5 h in 2038, against 15.5 h without the road layer: differences of +63.5 h [58.4, 68.7 h] and +82.9 h [76.0, 89.7 h], that is 5.1 and 6.3 times the baseline. Vehicles met 53 cross-owner conflicts a year in 2028, and the fleet's vehicles stood for 1,969 h a year in all waiting for a decision from Earth, plus 1,961 h in which the other vehicle of each conflict stood. In 2038 vehicles met 45 conflicts and stood for 2,527 h waiting for Earth: the poorer year cost more, in delay and in waiting, at today's fleet. Against the ATC mode on the same seeds the differences were +63.5 h [58.4, 68.9 h] and +82.9 h [76.0, 89.7 h]. At two times the fleet the uncoordinated delay was 8.1 times the baseline in 2028 and 10.3 in 2038, with 5,020 h of waiting for Earth in 2028; at four times it was 12.6 and 12.8 times the baseline, and at eight times 20.2 and 18.1 times.

5.5 Single-lane share, bay spacing and the lost-contact rule

With no single-lane road at all, the ATC mode's delay differed from the baseline by −0.03 h and the uncoordinated mode's by +0.1 h, the latter with 1.2 cross-owner conflicts a year. At a 5 % share the ATC mode differed from the baseline by +0.03 h and the uncoordinated mode by +63.7 h (5.1 times); at 40 %, with 96.8 km of one-lane road, the ATC mode differed by +0.28 h [−0.03, 0.58 h], +1.8 %, with 43 h of bay wait, and the uncoordinated mode by +79.7 h (6.1 times). Bays every 0.5 km instead of 1 km changed the ATC mode's delay by +0.06 h [−0.16, 0.27 h] and raised the segment traversals from 1,843 to 4,834; bays every 2 km changed it by 0.00 h, identically by construction, because at the default share no one-lane segment is longer than 1 km and the 2 km spacing places no bay. Under the strict reading of R7, in which a robot files a reservation only from a place the base's mast can see, the ATC mode's delay rose by 49.1 h [43.8, 54.8 h] over the default ATC mode on all 30 seeds, with 725 jobs a year that could not file, 37 conflicts referred to Earth, and vehicles standing 1,551 h a year in all waiting for it.

5.6 Encounters (H7)

Table 6 gives every type. Of the 300,000 encounters generated, 274,580 were driven (24,743 type-1 draws and 677 type-12 draws could not hold the vehicle on the slope). Eleven of the twelve types produced no hard violation in 264,580 driven encounters; their Bonferroni bounds run from 7.3×10^{-5} (type 1, following, effective n 75,257) to 1.2×10^{-3} (type 6, the landing closure, effective n 4,545). Type 9 (the nonconforming vehicle) had 58 violating encounters in 10,000 (effective n 6,115): 7 contacts, 58 below the required gap; an encounter may count under more than one kind. Its Clopper–Pearson upper limit is 8.1×10^{-3} on the unweighted count and 9.2×10^{-3} on its weighted violation share (0.61 %) at the effective n. Every one of them is in the dead-stop bin, where a fault stops the leader instantly; the slip and drift bins of the same type produced none (Section 6.1). H7 as registered, zero violations in every type, is therefore not supported: eleven types produced no violation, of which nine were also free of deadlocks; types 5 and 6 are clean of violations but not live (below). The importance weights reduce the effective n to 4,545 of 10,000 in type 6, 47,148 of 100,000 in type 7, 6,115 of 10,000 in type 9, 7,231 of 9,323 in type 12. Near misses, mostly a same-lane gap under twice d_{min}, occurred in 121,468 encounters (44 % of those driven); the smallest clearance in any type without a violation was 1.1 m. Deadlocks occurred in 386 of 10,000 type-5 encounters (3.9 %, the ring of one-lane segments with finite bays), 4 of 10,000 type-6 encounters (the landing closure) and in no other type. The four type-6 deadlocks are all at the 1,000 m zone radius, and a trace of each shows one common cause, a crossing that lies on the zone boundary (Section 6.1). A fault blocked 4,137 type-9 encounters before any rule was tested. The emergency brake fired in up to 52.7 % of a type's encounters (type 7, crew). The first main run, stopped after types 1–10 and part of type 11, had shown 37 violations in type 4, 3 violations in type 5, 82 violations in type 6, 58 violations in type 9 and none elsewhere. Of the violating encounters without a random fuzzer, which are deterministic and can be rerun as they were, 10 of 27 in type 4, 3 of 3 in type 5, 5 of 46 in type 6 were rerun after the fix and every one was clean (spot checks of the first few in batch order under a time budget, superseded by the full rerun at the registered n); 25 of 48 in type 9 were rerun after both fixes and every one still violated (25 below the gap, 4 contacts). The stopped run's files are archived and used for nothing else. The type-6 cell was run three times. Its first pass showed 10 R3 violations, vehicles that started inside their own stopping distance of a crossing tile, a start the protocol never faces, 10 gap counts between vehicles leaving a shallow crossing on different roads with 2.2 m or more between their bodies and no contact, and 201 encounters classed as deadlocks. The first was fixed in the builder and the second by narrowing the gap metric's same-lane test (same road and direction instead of 30°; the shield's leader test is unchanged). No violation count in any other type changes under the narrower test, and type 9's reproduced exactly; the near-miss counts of type 5 and the minimum-gap diagnostic of type 3 did change, so types 3 and 5 were rerun in full under the final metric. The second pass, with both corrections, had no violation and 202 encounters classed as deadlocks; a trace showed slow haulers correctly held at the closed zone, stopped just short of its line, that the stall check counted as waiting for a grant. The classifier was corrected (the deadlock flag and, in type 6, where the stall test ends an encounter; no violation changed) and types 5 and 6 rerun. The third pass is the one reported: type 6 fell from 202 deadlocks to the four above, and type 5's ring deadlocks stand at 386 (change 9 of [43]). Between the second and third passes an exploratory type-6 cell with an added request rule, registered in change 9 against a mechanism that the trace then withdrew, ran once (203 deadlocks against 202) and tested nothing; it is archived with the earlier passes and used for nothing else.

Table 6. E44, encounters per type: generated, driven and effective n, hard violations (by kind), the 95 % upper bound on the per-encounter violation probability with the Bonferroni correction over the twelve types, near misses, the smallest clearance, deadlocks, encounters blocked by a fault and the emergency-brake rate.

#	Type	n	driven	effective n	violations	bound (Bonferroni)	near misses	min margin (m)	deadlocks	blocked by fault	emergency brake
1	following	100,000	75,257	75,257	0	7.3×10^{-5}	69,925	5.2	0	0	0.0 %

#	Type	n	driven	effective n	violations	bound (Bonferroni)	near misses	min margin (m)	deadlocks	blocked by fault	emergency brake
2	head-on on a one-lane segment	10,000	10,000	10,000	0	5.5×10^{-4}	6,747	1.2	0	0	0.0 %
3	overtaking	10,000	10,000	10,000	0	5.5×10^{-4}	8,471	1.2	0	0	0.0 %
4	the junction	10,000	10,000	10,000	0	5.5×10^{-4}	221	1.2	0	0	0.0 %
5	the ring	10,000	10,000	10,000	0	5.5×10^{-4}	0	1.5	386	0	0.0 %
6	the landing closure	10,000	10,000	4,545	0	1.2×10^{-3}	137	1.1	4	0	0.0 %
7	crew	100,000	100,000	47,148	0	1.2×10^{-4}	4,023	4.1	0	0	52.7 %
8	radio dropout	10,000	10,000	10,000	0	5.5×10^{-4}	6,736	1.2	0	0	0.0 %
9	the nonconforming vehicle	10,000	10,000	6,115	58 (7 contact, 58 gap)	8.1×10^{-3}	4,525	−1.3	0	4,137	17.5 %
10	mixed owners and classes	10,000	10,000	10,000	0	5.5×10^{-4}	7,474	1.2	0	0	0.0 %
11	charger and pad contention	10,000	10,000	10,000	0	5.5×10^{-4}	9,825	1.2	0	0	0.0 %
12	planner fuzzing	10,000	9,323	7,231	0	7.6×10^{-4}	3,384	1.2	0	0	10.2 %

Bound: with no violation $1 - (0.05/12)^{1/n_{\text{eff}}}$; with violations the Clopper–Pearson upper limit at α/12 on the unweighted count (the limit on the weighted share at the effective n is given in the text). Effective $n = (\Sigma w)^2/\Sigma w^2$ under the importance weights of types 6, 7 and 9, which type 12 inherits from its type-7 draws. A negative minimum margin is a contact. Deadlocks are judged only in encounters without a random fuzzer.

5.7 The exhaustive check

The four layouts have 47,085 reachable states and 272,899 transitions in all; the largest, a pad visit with two oncoming vehicles and radio links, has 36,912 states. In every layout the search found no state that breaks a safety invariant, no deadlock state, no state from which the goal is unreachable and no livelock component; the longest shortest path to the goal is 41 steps of 8.5 s on 10 m cells. Each mutation produced its counterexample. Without R3's exit-slot check, 2 reachable states of the three-vehicle line are deadlocks and 39 cannot reach the goal (13 livelock components); the shortest deadlock trace is 12 steps. Without R5's entry check, 384 reachable states have a vehicle inside the landing zone while it is closed, the shortest reached in 14 steps. Without R7's hold, 198 reachable states break an invariant (138 in the segment without the reservation, 96 opposite directions in the segment; a state may break both), the shortest in 5 steps. The traces are stored beside the results. H7's second clause, a counterexample for each removed rule, holds. Development of the checker found four counterexamples before the final run, three in the checker itself (R6 checked as a state invariant rather than on the vehicle's motion, which flagged crew walking up to a stopped vehicle; no crew distance computed on the landing spur, 80 violating states in the first layout; bay slots counted per lane, which hid the R3 mutation) and one in the reading of a rule: with the bays fixed, the three-vehicle line deadlocked when a vehicle granted the segment with a free exit slot lost that slot to a vehicle that drove into the bay afterwards. R3's “reserved for it” therefore binds every other vehicle: a bay counts the slots reserved by segment holders as taken, in the checker and in the encounter simulator alike.

5.8 Mars (H9)

The Mars part ran E44's twelve types at $g = 3.71$ m/s^2 with the lunar vehicles, 44,000 encounters paired with the lunar encounters of the same seeds, and E43's two road modes at today's fleet with a Mars Earth link (Tables 7 and 8). The safety program allowed no violation in eleven of the twelve types; in the dead-stop type the

following vehicle closed below the required gap in 11 of 2,000 encounters, the same eleven encounters that violated on the Moon, and no vehicle touched another. With no violation the bound at 2,000 encounters is 2.7×10^{-3} with the Bonferroni correction (1.5×10^{-3} without), larger at the effective n of the weighted types; type 9's Clopper–Pearson limit is 1.2×10^{-2} with the correction (9.1×10^{-3} without), against 8.1×10^{-3} (7.2×10^{-3}) on the Moon at five times the sample. The ring deadlocked in 70 of 2,000 encounters (3.5 %, against 3.9 % on the Moon) and the landing closure twice in 2,000 (four in 10,000 on the Moon); no other type deadlocked. Of the 12,000 following draws, 8,961 (74.7 %) were on downhills steeper than 7.7°, the steepest the vehicles' 0.5 m/s^2 service brake holds at Mars gravity (Section 3.5), and were not driven; on the Moon the limit is 16.7–18.0° depending on the friction value, only the 20° draws are excluded, and 3,019 (25.2 %) of the same draws were not driven. No sloped encounter of any type was driven on Mars, and on the flat the braking term of R1 and R2 differs between the bodies by at most 0.01 m/s^2, so the driven Mars encounters repeat the lunar ones with the same seeds. For the 4,411 paired encounters that nominal planners drove to the end on both bodies, the rules delayed the vehicles by 67.0 s on Mars and 66.9 s on the Moon, a ratio of 1.00. The logistics model's ATC mode uses neither gravity nor the Earth link, and its 30 Mars runs equalled E43's ATC runs on every seed. The uncoordinated mode cost 10.4 h [7.8, 13.1 h], 11.1 h [8.5, 14.0 h] and 10.8 h [8.8, 12.8 h] of delay per delivery above the ATC mode at round trips of 6, 20 and 44 minutes without a relay, and 8.2 h, 8.7 h and 6.4 h with the relay (Table 8); at 20 minutes without a relay that is 0.72 of the ATC mode's 15.5 h. On the Moon the same difference was 63.5 h [58.4, 68.9 h] in 2028, 4.1 times the ATC mode's delay, and 82.9 h [76.0, 89.7 h] in 2038 (Table 5); the Mars cost is 0.18 of the Moon's 2028 cost. The Mars fleet referred 94 conflicts a year to Earth and waited 772 h for the decisions, 8.2 h per conflict; the lunar fleet referred 53 and waited 1,969 h, 37.2 h per conflict. H9's first clause, zero violations under Mars gravity, is supported on the flat in the same eleven types as H7, fails in type 9 for the same encounters, and is untested on slopes; its second, a similar rule cost, is tested only by the encounter delays, where it holds; its third, that Earth-dependent coordination loses more on Mars, is contradicted: it loses less in every Mars cell, with intervals that do not overlap the Moon's.

Table 7. E47 (Mars), encounters per type: generated, driven and effective n, hard violations (by kind), the 95 % upper bound on the per-encounter violation probability with the Bonferroni correction over the twelve types, near misses, the smallest clearance, deadlocks, encounters blocked by a fault and the emergency-brake rate.

#	Type	n	driven	effective n	violations	bound (Bonferroni)	near misses	min margin (m)	deadlocks	blocked by fault	emergency brake
1	following	12,000	3,039	3,039	0	1.8×10^{-3}	2,649	5.2	0	0	0.0 %
2	head-on on a one-lane segment	2,000	2,000	2,000	0	2.7×10^{-3}	1,368	1.2	0	0	0.0 %
3	overtaking	2,000	2,000	2,000	0	2.7×10^{-3}	1,669	1.2	0	0	0.0 %
4	the junction	2,000	2,000	2,000	0	2.7×10^{-3}	39	1.2	0	0	0.0 %
5	the ring	2,000	2,000	2,000	0	2.7×10^{-3}	0	1.5	70	0	0.0 %
6	the landing closure	2,000	2,000	906	0	6.0×10^{-3}	36	1.1	2	0	0.0 %
7	crew	12,000	12,000	5,664	0	9.7×10^{-4}	506	4.1	0	0	52.8 %
8	radio dropout	2,000	2,000	2,000	0	2.7×10^{-3}	1,365	1.2	0	0	0.0 %
9	the nonconforming vehicle	2,000	2,000	1,230	11 (11 gap)	1.2×10^{-2}	889	0.2	0	824	17.5 %
10	mixed owners and classes	2,000	2,000	2,000	0	2.7×10^{-3}	1,486	1.2	0	0	0.0 %
11	charger and pad contention	2,000	2,000	2,000	0	2.7×10^{-3}	1,964	1.2	0	0	0.0 %
12	planner fuzzing	2,000	1,600	1,194	0	4.6×10^{-3}	445	1.2	0	0	11.4 %

Bound: with no violation $1 - (0.05/12)^{1/n_{eff}}$; with violations the Clopper–Pearson upper limit at α/12 on the unweighted count (the limit on the weighted share at the effective n is given in the text). Effective $n = (\Sigma w)^2/\Sigma w^2$ under the importance weights of types 6, 7 and 9, which type 12 inherits from its type-7 draws. A negative minimum margin is a contact. Deadlocks are judged only in encounters without a random fuzzer.

Table 8. E47, the uncoordinated mode against the ATC mode at today's fleet: the paired difference in mean delay per delivery (h, 95 % bootstrap interval over seeds) and the same as a share of the ATC mode's delay, conflicts referred to Earth and the hours waited for Earth (fleet totals per run), on Mars by round trip and relay and on the Moon in the two E43 years (the rows of Table 5). The relay keeps the base in contact for the whole sol outside the conjunction and the storms. The ATC mode uses neither gravity nor the Earth link, so its Mars runs equal its lunar runs.

Body	Round trip (min)	Relay	Seeds	Uncoordinated − ATC (h)	95 % interval	Share of ATC	To Earth	Earth wait (h)
Mars	6	none	30	10.4	[7.8, 13.1]	0.67	95.6	748
Mars	6	areostationary	30	8.2	[6.0, 10.5]	0.53	90.0	479
Mars	20	none	30	11.1	[8.5, 14.0]	0.72	94.0	772
Mars	20	areostationary	30	8.7	[6.3, 11.2]	0.56	92.1	517
Mars	44	none	30	10.8	[8.8, 12.8]	0.69	96.5	836
Mars	44	areostationary	30	6.4	[4.2, 8.7]	0.41	97.9	534
Moon, 2028	2.6 s	–	30	63.5	[58.4, 68.9]	4.09	52.9	1,969
Moon, 2038	2.6 s	–	30	82.9	[76.0, 89.7]	5.33	44.6	2,527

5.9 What did not work

The eight-times cells ran at 60 days on ten seeds (40 runs) after the dispatch fix of Section 4.6; without it they could not run, and no 30-day row was used. Every cell ran all its seeds. Of the 86 phase-5 checks of Section 3.8 tagged to this paper, 7 failed: D10 pairing A_EARTH plan_ahead (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_EARTH plan_ahead (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing A_EARTH strict (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_EARTH strict (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing A_MOON (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): A_MOON (E40, 10 % load, 3eq)), the condition did not hold; D10 pairing B_EARTH strict (CRN pairing reduces variance against O_ORACLE (correlation's 95 % CI above zero): B_EARTH strict (E40, 10 % load, 3eq)), the condition did not hold; S1 (c) type 9, non-conforming (Zero separation violations with non-conforming vehicles (type 9: slip, drift, dead stop)), the condition did not hold; S2 unholdable slope flagged (Injected unsafe proposals refused: unholdable slope flagged), the condition did not hold; S5 encounter deadlocks (Zero deadlocks in the encounter layer (E44)), the condition did not hold. The other 78 that ran passed; 1 was not run: T5 (c) node conservation (not applicable: no node occupancy in the engine; site capacity is the apron slot rule (T5 (b))). Of the failures, S1 (c) type 9, non-conforming (58 violations) and S5 encounter deadlocks (390 encounters) are the findings of Section 5.6, type 9's violations beyond the leader-braking assumption and the deadlocks of change 9. The S2 row flags the shield's one accepted unsafe case: of 1,000 injected proposals to brake on a slope that friction cannot hold, 990 were refused and 10 accepted, all at near-standstill, a case that R2 keeps every run from reaching (Section 7). After every experiment had run, a verification check found that cargo ids were not unique: 9–18 of about 2,970 items a year reused an earlier id, in every regime (change 10 of the pre-registration). Ids enter the simulator's decisions only through a planning memo within one dispatch call. The fix was tested by rerunning 48 full years (E40, E43: every regime or mode, seeds 1–3), of which 48 gave outcomes identical item by item, compared by content without ids, 9 of them in this paper's experiment; reused ids fell from 670 to 0. No experiment was rerun and no reported number changes; trace and scene files written before the fix still carry the duplicate ids. Nothing else is withheld: every E43 cell that ran is in Table 4 and every E47 cell in Tables 7 and 8; the second Mars geometry of Section 3.7 was planned and not run.

6. Discussion

6.1 What the results say about the question

The question had three parts: whether a small shared rule set keeps robots of several owners safe against any planner, what it costs, and whether it holds on Mars. On cost the answer is clear and, we think, the most useful number in the paper. At today's planned fleet the rules cost 0.1 % of delay per delivery, an amount that no operator would notice, and the whole fleet lost 24 h a year at passing bays. That is because the base's roads are long and its vehicles few: at today's fleet the whole fleet meets an oncoming vehicle on a one-lane segment about 1.4 times a week (73 conflicts a year), and a wait at a bay is minutes against deliveries measured in hours. The cost grows with the fleet, to 7.9 % at four times the fleet and demand, which is the congestion the warehouse literature predicts [6, 17], and the price of safety at eight times is 41.7 % of delay per delivery. The baseline delay itself fell as the fleet grew, because the pooled dispatcher had more vehicles to choose from for each item, so the rules' cost is a share of a smaller number at the larger fleets. The rules as implemented in the logistics model never deadlocked, in 650 runs at up to eight times the fleet; that is the exit-bay construction of R3 doing what the theory says it does [19, 23]. That zero is a property of the logistics model rather than of the rule: its bays are unbounded and its reservation table is filled one job at a time (Section 3.4), so no circular wait can form there, and the count is a check of the code. With finite bays the exit-bay rule is not deadlock-free on a ring of single-lane segments, as the encounter simulator's ring type shows (Section 5.6), and whether this base has such a ring the 5 m route samples cannot settle: merged to road nodes within 40 m they show one candidate ring of 419 m at the default share, one at a 5 % share and 14 at 40 %, but the merge can itself join neighbouring roads, and before merging the 5 loops at the default share are one road sampled twice by different legs. A surveyed centreline map would settle it; the samples do not.

The second number is the one we would put in front of a policymaker. The same rules, with each owner keeping its own reservation table and every cross-owner conflict settled by hand from Earth, cost 5.1 times the baseline delay at today's fleet, with its vehicles standing 1,969 h a year in all waiting for a decision that could only be made while the base saw Earth. The rules did not change between the two modes; the place of the decision did. A conflict on a one-lane road is settled in milliseconds by a shared table on the Moon and in hours by two control rooms on Earth, and for about eleven days in every month it is not settled at all [42]. The strict reading of the lost-contact rule tells the same story from the other side: a robot that may not file a reservation until the base's mast sees it behaves like an uncoordinated one for the first leg of every job, and under that reading the ATC mode's delay per delivery differed from the default's by 49.1 h. The rule that matters most for cost is therefore not any of the ten in Table 2 but the one about who may grant a reservation.

On safety, the exhaustive check says that the rules are consistent on every layout the checker can hold, 47,085 states of up to three vehicles on one segment with its bays, a pad and a crew member, and that each of the three clauses we removed is load-bearing: without it the search finds an unsafe or a stuck state within a few steps. It says nothing about larger layouts, which is where the simulator takes over. The simulator's one failure is where the derivation says it should be. R1's gap is computed for a leader that brakes at b_{max}, and a vehicle that stops dead, a fault rather than a manoeuvre, lies outside that model [14], so a same-direction follower already inside a one-lane segment can close below the gap and, in 7 of 10,000 encounters, touch. A dead-stop margin, the follower's own stopping distance at its current speed, would close the case at a cost in segment capacity; we did not run it, and the count stands as a limit of braking-based separation. The ring deadlocks of type 5 (3.9 % of its encounters) are the property noted before the main run [43]: the exit-bay rule is not deadlock-free on a ring of

one-lane segments with finite bays, and the logistics model's zero count comes from its unbounded bays; the count stood when the stall classifier was corrected (change 9 of [43]). The four deadlocks of type 6 are a gap in the rules: a crossing that lies on a landing zone's boundary can be held by a vehicle stopped at the zone line, and the vehicle leaving the closing zone through it is blocked. R3 keeps a vehicle out of a segment it cannot leave into a bay, but no rule keeps it out of a junction whose far side is a hold point. The repair is a one-line extension of R3's exit check to junctions; we did not run it, and the count stands as found. On Mars, the safety program behaved as on the Moon, the dead-stop exception included, but that is weaker evidence than it looks: the only ground the lunar brakes hold at Mars gravity is nearly flat, and on the flat the braking term is the same on both bodies, so the Mars encounters that were driven repeat the lunar ones; the rules are untested on the slopes where Mars gravity acts. The same holds for the rules' cost, measured only by the paired encounter delays. The Earth-dependent mode lost about one-sixth of what it loses on the Moon, 11.1 h against 63.5 h, the opposite of H9's third clause. The reason is in the Earth windows, not the round trip: the three round trips gave the same cost within their intervals, and the relay lowered the mean at each, while a Mars base, as we modelled it, sees Earth for half of every sol (Assumed, Section 7) and the polar base's computed windows [42] leave it blind for days, so a referred conflict waited 8.2 h for a decision on Mars and 37.2 h on the Moon. The reversal therefore holds for any Mars site whose horizon returns Earth every sol, and would not for one that does not. The Mars fleet referred more conflicts (94 against 53) for the same demand and almost the same deliveries (2,932 against 2,928 items); we have not traced the cause. The one Mars-specific finding is about hardware. The guaranteed braking of Section 3.5 is the smaller of the service brake and the friction limit, less the downhill component: on the Moon the friction term binds at $\mu = 0.3$ and the 0.5 m/s^2 service brake above it, so the holdable downhill is 16.7–18.0°; on Mars the service brake binds at every friction value, so braking does not grow with gravity while the slope term does, and the holdable downhill falls to 7.7°. Of the following draws, 75 % were on steeper ground, against 25 % on the Moon. R2 gives a zero admissible speed on such a slope, so the safety program refuses the slope; it has no rule that says a vehicle is under-braked for the body it is on, and vehicles for Mars need a service brake sized for Mars gravity.

6.2 Comparison with earlier studies

Our cost of safety is smaller than the warehouse and intersection literature reports, for a reason that is in the numbers rather than in the method. Hönig and colleagues' execution layer cost nothing in unnecessary wait and beat synchronous execution by 2.5 times in station use [5]; Dresner and Stone's reservations added under 0.1 s to travel time at densities that saturated every traffic light [7]; lifelong warehouse planners lose throughput only past a density peak of hundreds of agents on a few thousand cells [6, 17]. Our base has eight vehicles on 242 km of road, and the density ladder of Section 5.3 shows the same shape, with the cost becoming visible between two and four times today's fleet. The uncoordinated mode's cost has a closer analogue in drone traffic management, where NASA's tests found operational impact from downed servers and misimplemented specifications rather than from latency [9]; on the Moon the equivalent of a downed server is the horizon, and it is down for eleven days at a time. Our deadlock result agrees with the theory of well-formed layouts [19, 20] and with Dresner and Stone's observation that coarse reservations deadlock [7]: our reservations are whole segments, which is coarse, but they end at bays, which is what the theory requires. Our per-type bounds are the zero-numerator bounds of Hanley and Lippman-Hand [33]: with no violation in n encounters the 95 % upper limit is close to 3/n, 3.0×10^{-5} at 100,000 and 3.0×10^{-4} at 10,000, larger at the smaller effective n of the weighted types, and they hold for the generated distribution of each type, not for the base's actual traffic. Zhao and colleagues reach rarer events by importance sampling on the lane-change parameters [35]; we used the same device on three types (the

exit distance, the crew's emergence time and the dead-stop time), which cost effective sample size (Section 5.6) and found the one violating case, type 9, where the boosted region was.

6.3 Implications

For the agreements that the Artemis Accords leave to the parties [1], the implication is that the rules themselves are cheap and the sharing of a reservation table is where the value lies. A base that adopts rules like Table 2 but lets each company keep its own table, with conflicts escalated to Earth, keeps the safety of the rules and pays 5.1 times the delay; one that shares the table on the Moon pays 0.1 %. The shared table does not require a shared planner, a shared dispatcher or a shared owner, only that every vehicle run the same small program and accept a grant from the base computer, which is the federated structure of drone traffic management [8, 25] with the server moved to the Moon. For NASA's policy question of when central coordination becomes necessary [2], the simulation says: as soon as there is a one-lane road and two owners. For designers, the number that drives cost is the share of one-lane road, which the 5 m terrain cannot give; a survey of the real corridors at rover scale, or a graded road, would settle what we had to assume.

6.4 Limitations

The findings hold for one base, one route network and one demand model, with the single-lane share assumed; the road layer acts at a 100 m grain and leaves separation within a segment to the encounter simulator; the uncoordinated mode's accounting understates its cost; the encounter bounds hold for the generated distribution and the proof for the discrete model; and no sensor is modelled; and Mars is the Moon's roads with Mars parameters, with its encounters repeating the lunar ones on the flat. Section 7 states each in full.

6.5 Future work

Five pieces follow directly. A survey of corridor widths at rover scale, from higher-resolution imagery or a graded road plan, would replace the assumed single-lane share. The safety program should be run on the flight-class computer the companion paper sizes [45], against recorded traffic from this model. The lunar uncoordinated mode should be given the relay the Mars part had and the plan-ahead option of [44], to see how much of its cost they recover. The encounter simulator should be extended to a lander in flight and to the dust a landing raises, which the rules here treat only as a closed zone. And a Mars base on its own terrain, which needs a reader for the HiRISE tiles, a site and a route network, would test the rules on the slopes where Mars gravity acts.

7. Limitations

The single-lane share is assumed. At the rovers' slope limit the 5 m terrain shows no corridor narrower than two vehicles, so every one-lane segment in E43 is a choice: a share of the route samples, the narrowest at a 10° criterion first. The sweep from 0 to 40 % brackets the plausible range, and the ATC mode's cost stayed under 3.7 % across it, but the absolute waits depend on where the real pinch points are.

One base, one network, one demand. The Connecting Ridge base of [41, 42], its routes and the demand model of [40] are the only ones simulated. The fleet multiples change the number of vehicles and items, not the road network, so the density ladder is a ladder of traffic on fixed roads.

The road layer has a 100 m grain. In the logistics model vehicles reserve whole segments at job commit and following vehicles are not separated; R1, R2 and R9 act only in the encounter simulator. A vehicle that would be

delayed within a segment by a slower one ahead is not delayed in E43, which understates the ATC mode's cost by an amount the encounter simulator bounds but does not price.

The uncoordinated mode's cost is understated. When a cross-owner conflict stops two vehicles, the other vehicle's stop lengthens its busy time but deliveries it had already recorded are not moved, and the hand-resolution time of two hours is assumed. Both choices favour the uncoordinated mode.

The rules are checked, not validated. The exhaustive check covers a small discrete model, the encounter bounds hold for the twelve generated types and their parameter ranges, and neither covers real sensors, navigation error beyond the conformance margin, or a vehicle that breaks the rules on purpose. Owners who misreport are the subject of [45].

The safety program is one implementation. What is checked is our code; a vehicle maker's implementation of the same rules would need the same checks against its own code.

The shield accepts one unsafe case no run reaches. The phase-5 check S2 injected 1,000 proposals to brake on a slope steeper than friction can hold and the shield refused 990; the 10 it accepted were vehicles at near-standstill, because the shield's kinematics take a stopped vehicle to stay stopped. R2 never admits a vehicle to such a slope, so no run of E43, E44 or E47 reaches the case, and the shield was not changed after the runs; a flight implementation should flag it.

A hand-stopped vehicle's plotted position runs early. When Earth resolves a conflict in the uncoordinated mode, the engine shifts the reservation and the busy time of the vehicle that goes first but not its leg plan, so that vehicle's plotted position runs ahead of its reservation by the hand delay until the job ends. Delay, deliveries and waits are computed from busy times and reservations, which are shifted, so no reported number depends on it; the plotted positions feed only the mesh radio, which the road runs of this paper do not use. The defect was found by the trace export after the runs and was not corrected in the engine; the exported trace follows the reservation.

Horizons and seeds differ by fleet size. The four- and eight-times cells run 90 and 60 days on ten seeds, not a year on thirty, so their intervals are wider and their delays are not directly comparable with the year-long cells' absolute values; the paired differences within a cell size are the result. The horizons were fixed in the design of E43 before the dispatch fix of Section 4.6 and kept so that the rerun matched the archived rows; after that fix they are a choice of the design, not a limit of compute.

Mars is the Moon's roads with Mars parameters. The Mars part runs the same base, roads, routes and demand with Mars gravity, a Mars Earth link and the lunar vehicles, and its encounters repeat the lunar ones on the flat (Section 6.1); a Mars site with its own terrain, roads and vehicles was planned and not run, because the terrain reader is written for the lunar polar tiles; the share of each sol with Earth in view, the conjunction date and the dust storms are assumed, and the simulated year holds a conjunction, which comes once in about 26 months, a conservative choice; and at 2,000 encounters per type the Mars bounds are five times the Moon's for the same type, 2.7×10^{-3} for the unweighted types with the Bonferroni correction and up to 6.0×10^{-3} for the weighted types (Table 7).

8. Conclusion

We asked whether shared traffic rules keep robots of several companies safe at a lunar base, what they cost and whether they carry to Mars. We wrote ten rules, each parameter sourced or marked assumed, and tested one

safety program three ways in simulation. The exhaustive check found no unsafe or stuck state in 47,085 reachable states; the simulator found no violation in eleven of twelve encounter types, one in the twelfth, a vehicle that stops dead inside a one-lane segment; the ring of one-lane segments with finite bays deadlocked in 3.9 % of its encounters, and the landing closure four times in 10,000 at a crossing on the zone boundary. At today's fleet the rules cost 0.1 % of delay per delivery, 7.9 % at four times, and never deadlocked. Settling every cross-owner conflict from Earth cost 5.1 times the baseline delay. On Mars the rules behaved as on the Moon and referral to Earth cost about one-sixth as much; at Mars gravity the lunar vehicles' brakes hold no downhill steeper than 7.7°.

For practice, the value lies in the shared reservation table: a base whose robots settle conflicts themselves pays almost nothing for safety; one that keeps the decision on Earth pays in every blackout. For research, the one-lane share of a real base's roads is what this work most needs; the results hold for one base and a checked, not validated, model. What the Artemis Accords leave open can be written down, checked and priced.

Data and code availability

The simulation engine, the road layer, the safety program, the encounter simulator, the model checker, the experiment runner, every results file this paper quotes and the script that builds the paper from them are in the repository at github.com/azizjonrahimov/Lunar-Haul (directories `lunarsim/atc`, `experiments/phase5`, `results` and `paper/p2`). The ranked single-lane places with coordinates are `results/ATC_roads_narrow.json`, the corridor analysis `results/ATC_roads_corridor.json`, and the design of E43 with its dated changes `data/scenarios/E43-design.json`. The pre-registration with its dated changes is `paper/atc/PREREG.md` [43]. The live site at azizjonrahimov.github.io/Lunar-Haul shows the base, the vehicles, an encounter sandbox and the results.

Author contributions

A.R. set the questions, the scope of the three papers and the standards of evidence, directed the design of the rules, the simulation and the experiments, reviewed the hypotheses, the results and the text, and is responsible for the content.

Use of AI tools

Large-language-model assistants (Anthropic's Claude, used through Claude Code) wrote the simulation code, the road layer, the safety program, the experiment runner and the analysis scripts, ran the experiments, drafted and revised the text and compiled the reference list, all under the author's direction and review. Every number in the paper is written by the build script from the results files, and each sentence that states a direction or a verdict is checked by that script against the data; no number was typed from an assistant's output. The references were collected from published sources and the repository's research notes; entries not yet checked against the publisher's record are marked in the notes and will be checked before submission. The author takes responsibility for the whole.

Funding and competing interests

No external funding was received. The author founded Roatify, which proposes a traffic and dispatch network for lunar vehicles of the kind simulated here. The pre-registration, the reported assumptions and the build-script checks are the safeguards offered against that interest.

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